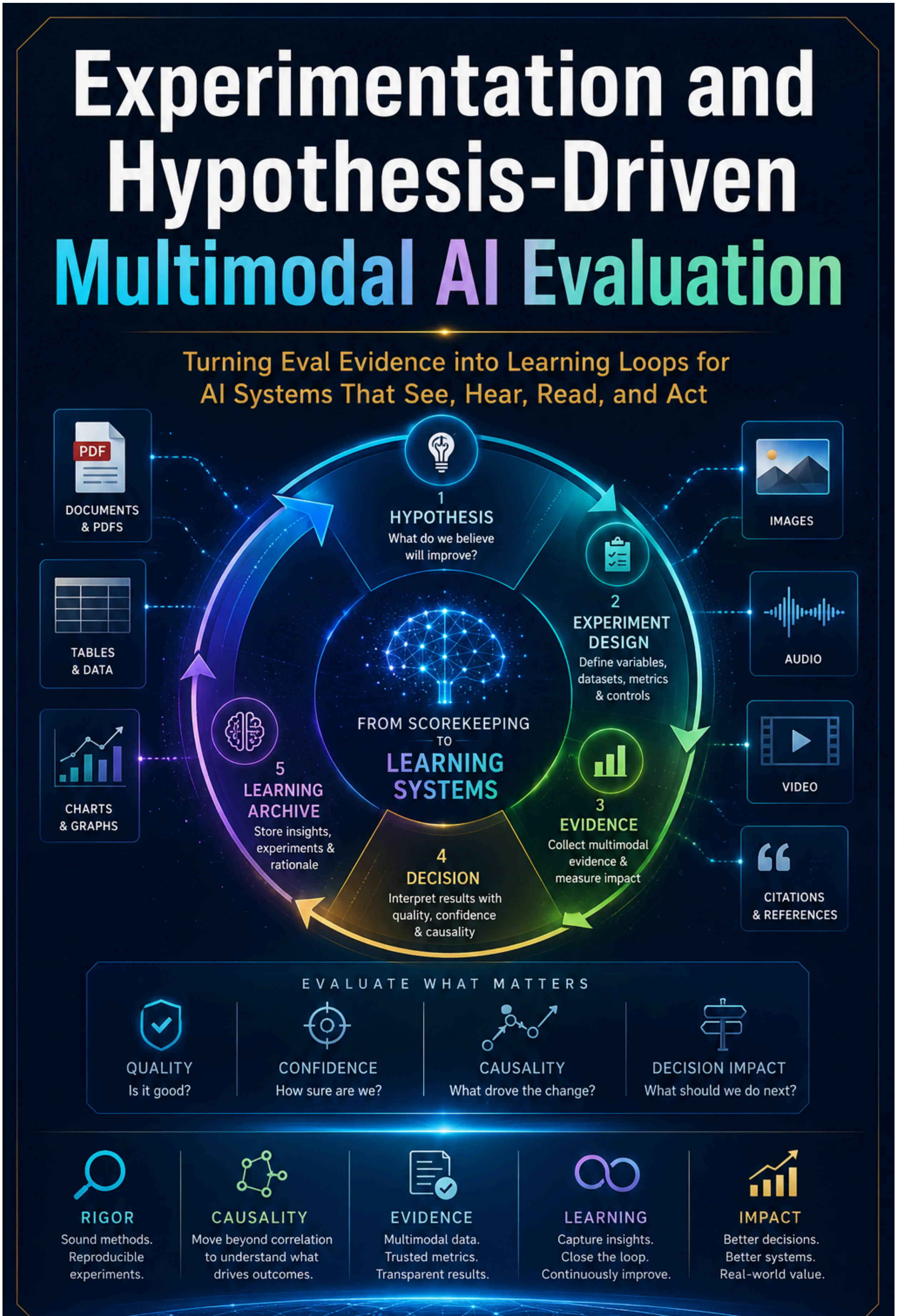

Experimentation and Hypothesis-Driven
Multimodal AI Evaluation
Turning Eval Evidence into Learning Loops for AI Systems That See, Hear, Read, and Act
PDF
DOCUMENTS & PDFS
TABLES & DATA
CHARTS & GRAPHS
IMAGES
AUDIO
VIDEO
CITATIONS & REFERENCES
1 HYPOTHESIS
What do we believe will improve?
2 EXPERIMENT DESIGN
Define variables, datasets, metrics & controls
3 EVIDENCE
Collect multimodal evidence & measure impact
4 DECISION
Interpret results with quality, confidence & causality
5 LEARNING ARCHIVE
Store insights, experiments & rationale
FROM SCOREKEEPING TO LEARNING SYSTEMS
EVALUATE WHAT MATTERS
QUALITY
Is it good?
CONFIDENCE
How sure are we?
CAUSALITY
What drove the change?
DECISION IMPACT
What should we do next?
RIGOR
Sound methods. Reproducible experiments.
CAUSALITY
Move beyond correlation to understand what drives outcomes.
EVIDENCE
Multimodal data. Trusted metrics. Transparent results.
LEARNING
Capture insights. Close the loop. Continuously improve.
IMPACT
Better decisions. Better systems. Real-world value.

Experimentation and Hypothesis-Driven Multimodal AI Evaluation

Turning Eval Evidence into Learning Loops for AI Systems That See, Hear, Read, and Act

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Multimodal AI Evaluation Should Not Be a Scoreboard

A team reviews its latest release. The Multimodal document intelligence agent — V2 — has improved Faithfulness from 84% to 88%. Cross-Modal Consistency has moved from 0.82 to 0.86. The dashboard is green. The release slide reads "Recommend ramp."

Then someone in the room asks the only question that matters.

"Why did the score move?"

Silence.

Was it the new OCR engine?

The new page segmentation?

The new table extractor?

The new chart parser?

The new retriever?

The stronger base model?

Better prompt instructions?

A more lenient Judge?

A re-curated eval set?

A shift in production traffic distribution?

A combination of all ten landed in the same release branch?

Nobody could answer.

This is the quiet failure mode of modern Multimodal AI evaluation. The team has movement, but no causal story. A score, but no learning. A green arrow, but no map for what to try next.

The argument of this document is that Multimodal AI evaluation should stop being a scoreboard and start being a learning system.

The discipline that turns one into the other is **experimentation** — and the central technique inside experimentation is **hypothesis-driven evaluation**.

The central claim is simple:

Evaluation measures quality.

Statistics tells us whether we can trust the measurement.

Experimentation tells us what actually changes quality.

A dashboard tells a team what moved. A statistical report tells them whether the movement is trustworthy. An experiment tells them *why* it moved, *for whom* it moved, *what risks moved with it*, and *what to do next*.

Why Experimentation Matters After Statistical Evaluation

This document is a direct extension of the earlier three-volume work on the *Statistical Aspects of Multimodal AI Evaluations*. That work covered how teams read evidence responsibly, produce evidence defensibly, and institutionalize evidence through EvalOps, Quality Contracts, Statistical Debt Registers, and the Risk-Aware Operating Loop.

Statistics and experimentation answer different questions.

Statistical aspects answer:

- How reliable is this measurement?
- How large is the uncertainty?
- Are the results statistically and practically meaningful?
- Is the judge calibrated?
- Were the results stratified by slice?
- Are we comparing fairly?

Experimentation aspects answer:

- What change caused the improvement or regression?
- Which component moved the metric?
- Which slice benefited and which slice was harmed?

- Which guardrail was stressed?
- What should we try next?
- What learning should be added to the organization's knowledge base?

The two layers sit on top of each other. A team can produce statistically sound measurements and still fail to learn anything, if it never forms a clear hypothesis about *why* something should improve.

Statistical evaluation helps teams avoid false confidence. Experimentation helps teams avoid false learning.

The cost of skipping the experimentation layer is not just slower learning. It is invisible regression. It is shipped lift that came from a more lenient judge. It is "improved" cross-modal consistency that quietly destroyed multilingual performance. It is a Faithfulness gain that masks a Factuality drop in numerical summaries.

Evaluation vs. Statistical Evaluation vs. Experimentation

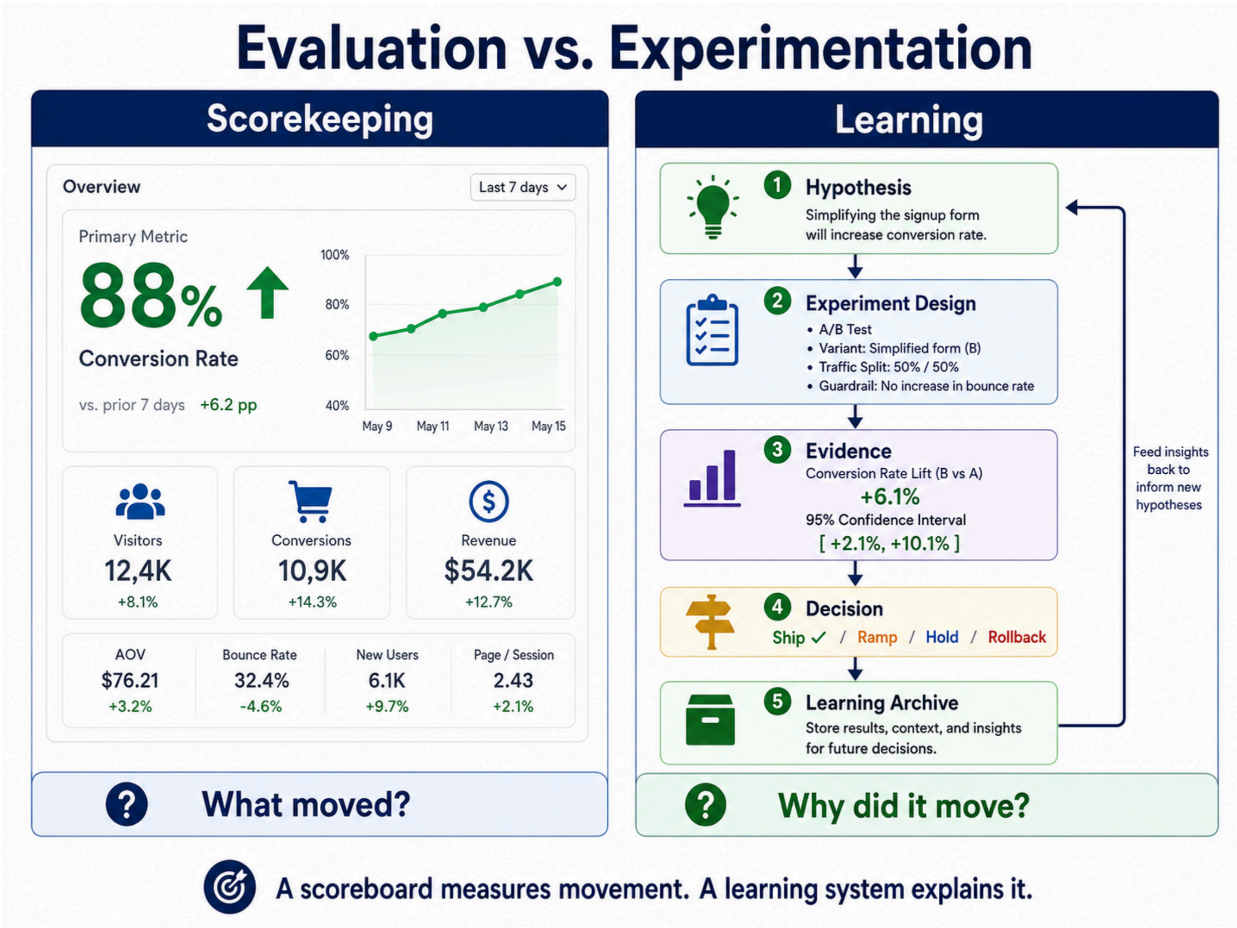
These layers are often conflated. They should not be.

Layer	Core Question	Output
Evaluation	How good is the system?	Score, error profile, quality signal
Statistical Evaluation	How much should we trust the score?	Confidence intervals, sample size, judge reliability, stratified evidence
Experimentation	What caused the score to move?	Validated or rejected hypothesis
Hypothesis-Driven Evaluation	What do we believe will improve, why, and how will we test it?	Pre-registered learning loop
EvalOps	How do we make this repeatable?	Infrastructure, workflow, governance
Governance	What decision should we follow?	Ship, ramp, hold, rollback, redesign

The distinction worth memorizing:

An eval run produces a score. A statistical report qualifies the score. An experiment tests a belief. A learning system accumulates validated beliefs over time.

Evaluation without experimentation produces dashboards. Experimentation turns those dashboards into learning.



Defining the Discipline

A working definition of the broader discipline:

Experimentation aspects of Multimodal AI evaluations refer to the disciplined design, execution, analysis, and learning capture of controlled changes to Multimodal AI systems in order to understand what improves or degrades quality, safety, reliability, user outcomes, and decision readiness.

The central technique inside that discipline is hypothesis-driven evaluation:

Hypothesis-driven Multimodal AI evaluation is the disciplined practice of testing whether a specific intervention improves a specific quality contract for a specific

modality, task, workflow, user segment, or production slice by a meaningful amount, without violating pre-defined guardrails.

The canonical hypothesis format follows directly from the definition:

*We believe **[intervention]** will improve **[quality contract]** for **[target slice / task / modality]** by **[minimum meaningful effect]**, without degrading **[guardrail metrics]**. We will decide **[ship / ramp / hold / rollback / investigate]** if **[decision rule]** is met.*

A worked example:

We believe adding table-aware chunking will improve Faithfulness on scanned financial PDFs by at least 4 percentage points, without reducing Ingestion Fidelity below 98%, increasing latency beyond 3 seconds, or increasing factuality errors in numerical summaries. If the lift clears the pre-registered threshold across 500 paired examples and no zero-tolerance failures occur, we will ramp to 25% of traffic.

Notice what the format does. It names the change. It names the contract. It names the slice. It names the threshold. It names the guardrails. It names the decision. There is nowhere for ambiguity to hide.

What Makes a Good Hypothesis

A good Multimodal AI evaluation hypothesis is **specific** (names the intervention precisely), **causal** (states why the change should work), **measurable** (maps to metrics and Quality Contracts), **sliced** (identifies where the effect should appear), **thresholded** (defines what improvement is meaningful), **guardrailed** (names what must not regress), and **decision-bound** (states what action follows the evidence).

Weak Statement	Strong Hypothesis
Improve chart understanding	We believe chart-structure extraction will improve Cross-Modal Consistency on chart-heavy financial PDFs by at least 5 points without reducing Faithfulness or increasing latency beyond 3 seconds.
Make the meeting agent better	We believe speaker diarization will reduce speaker-attribution errors by at least 5 points on multi-speaker meetings without increasing summarization latency beyond 2 seconds.
Reduce hallucinations	We believe citation-constrained generation will reduce unsupported claims by 4 points on document QA tasks without increasing refusal rate above 3%.

Improve safety	We believe tool-call policy constraints will reduce prompt-injection success rate below 1% without reducing task completion by more than 2 points.

The weak versions are wishes. The strong versions are tests that can be lost.

**The purpose of hypothesis-driven evaluation is not to prove that a model is better.
It is to decide what to do next with less self-deception.**

A note on skeptical hypotheses. The format works equally well — and is often most valuable — for the case the team *expects to fail*. "We believe replacing the current retriever with the trendy new vendor will *not* meaningfully improve Faithfulness on our domain" is a legitimate, decision-bound hypothesis. Pre-registering skeptical hypotheses is the cheapest defense against confirmation bias on shiny new techniques and against the slow organizational drift toward whatever vendor's demo looked most impressive last quarter.

The CLAIM Framework

To make hypothesis writing repeatable, this document proposes a five-part template called **CLAIM**.

CLAIM: A Hypothesis-Driven Multimodal Evaluation Template

Letter	Meaning	Question
C	Change	What are we changing?
L	Logic	Why should this change improve the system?
A	Assessment	How will we measure the effect?
I	Impact	What improvement is meaningful?
M	Mitigation	What risks or regressions must be blocked?

Worked example — speaker diarization for meeting summaries.

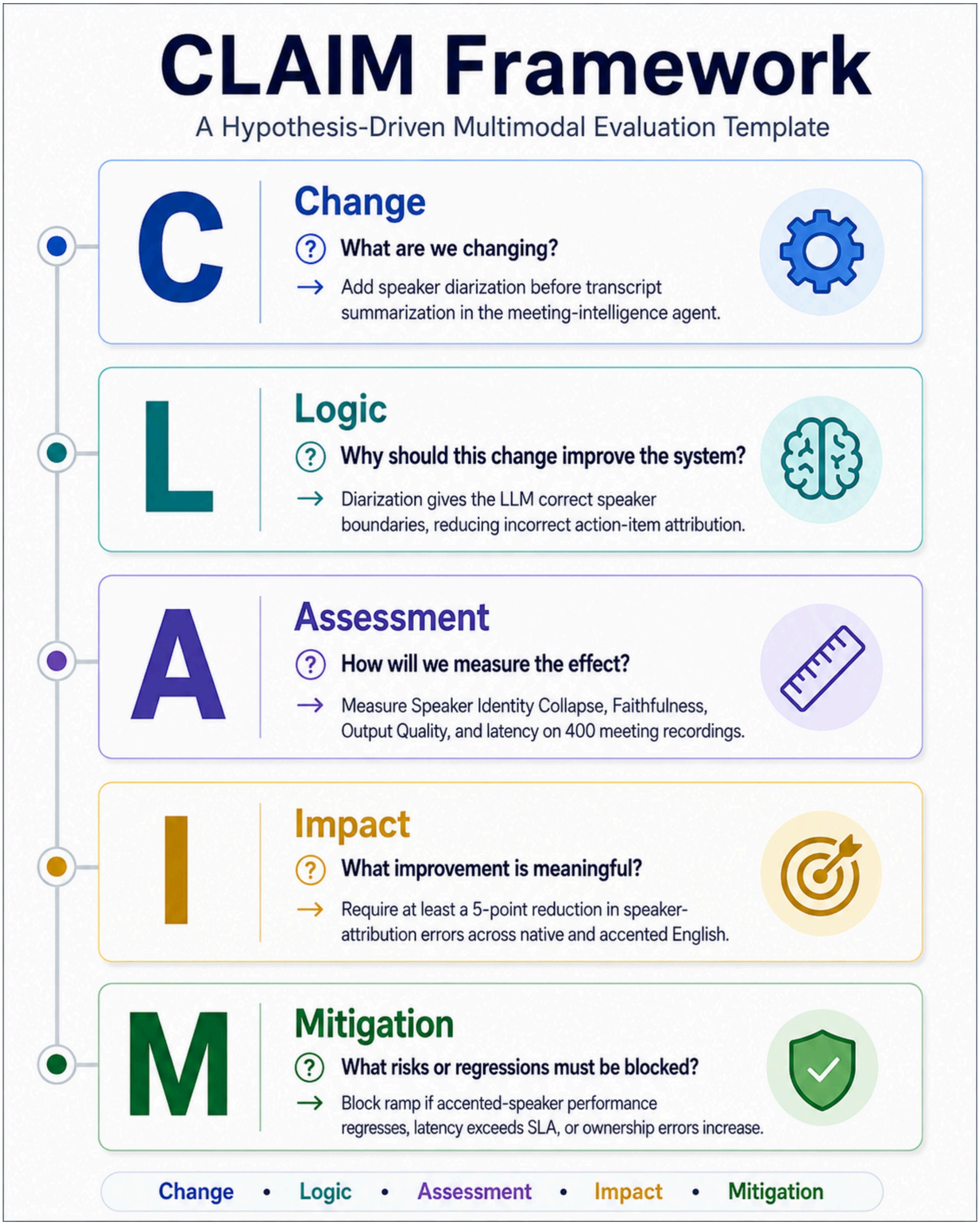
Scenario. A Multimodal meeting assistant confuses speakers in long meetings, producing inaccurate summaries and misattributed action items.

- **C — Change:** Add speaker diarization before transcript summarization in the meeting-intelligence agent.

- **L — Logic:** Speaker-identity errors are leaking into summaries and producing incorrect action-item attribution. Diarizing before summarization should give the LLM correct speaker boundaries to anchor attribution.
- **A — Assessment:** Measure Speaker Identity Collapse, Faithfulness, Output Quality (rubric), and end-to-end latency on 400 paired meeting recordings spanning native and accented English.
- **I — Impact:** Require at least a 5-point reduction in speaker-attribution errors, sustained across both native and accented slices.
- **M — Mitigation:** Block ramp if accented-speaker performance regresses, end-to-end latency exceeds the SLA, or action-item ownership errors increase in any slice.

Why CLAIM works:

- It makes the intervention explicit (C).
- It forces causal reasoning *before* evidence is collected (L).
- It connects evaluation to measurement (A).
- It defines what counts as meaningful (I).
- It prevents one contract from improving while another is silently harmed (M).


CLAIM Framework
A Hypothesis-Driven Multimodal Evaluation Template
C
Change
What are we changing?
Add speaker diarization before transcript summarization in the meeting-intelligence agent.
L
Logic
Why should this change improve the system?
Diarization gives the LLM correct speaker boundaries, reducing incorrect action-item attribution.
A
Assessment
How will we measure the effect?
Measure Speaker Identity Collapse, Faithfulness, Output Quality, and latency on 400 meeting recordings.
I
Impact
What improvement is meaningful?
Require at least a 5-point reduction in speaker-attribution errors across native and accented English.
M
Mitigation
What risks or regressions must be blocked?
Block ramp if accented-speaker performance regresses, latency exceeds SLA, or ownership errors increase.
Change • Logic • Assessment • Impact • Mitigation

The Running Example

The rest of this document uses a single running example to make the abstractions concrete.

System. A Multimodal document intelligence agent that processes scanned financial reports — annual reports, 10-Ks, fund factsheets, broker decks. The documents contain dense tables, charts, captions, footnotes, and numerical summaries. The agent answers analyst questions with grounded citations.

Intervention under test. Two coupled changes — *table-aware chunking* (preserving table structure during ingestion) and *chart-structure extraction* (parsing charts into structured key-value form before retrieval).

Why this is interesting. Both changes touch ingestion, both could improve Faithfulness and Cross-Modal Consistency, both could regress latency and Ingestion Fidelity if misimplemented, and the slices that benefit may be different (scanned vs. born-digital, dense-table vs. chart-heavy). It is exactly the kind of release where a headline lift can conceal a slice-level regression — which is what makes it useful as a teaching example.

We will return to this example for hypothesis writing, experimental design, decision rules, the T-19 brief, and the Learning Archive.

A Taxonomy of Hypotheses

Not all hypotheses are the same shape. Teams often conflate product, model, risk, and measurement hypotheses, which leads to muddled experiments.

In practice, a single intervention usually generates *multiple* hypotheses across types. Table-aware chunking, for instance, simultaneously creates a Quality Hypothesis (Faithfulness lift), a Slice Hypothesis (effect on dense-table documents), an Architecture Hypothesis (chunker contribution beyond OCR-only baseline), and a Judge Hypothesis (whether the judge can reliably grade table-grounded answers). Naming all of them at the outset keeps the experiment honest.

Six useful types:

1. **Quality Hypothesis** — A change improves a Quality Contract. *Example:* Table-aware chunking improves Faithfulness on scanned financial PDFs.

2. **Risk Hypothesis** — A change reduces a named harm. *Example:* Prompt-injection filtering reduces unsafe tool calls in agentic browsing.

3. **Slice Hypothesis** — A change improves a specific segment. *Example:* ASR denoising improves performance on noisy call-center audio without harming studio-quality recordings.

4. **Architecture Hypothesis** — A component contributes meaningful value beyond a simpler baseline. *Example:* The chart parser improves Cross-Modal Consistency beyond a prompt-only baseline.

5. **Judge Hypothesis** — A measurement instrument is reliable enough to depend on. *Example:* The LLM judge agrees with human raters at $\kappa \geq 0.75$ for cross-modal consistency on financial charts.

6. **UX Hypothesis** — A presentation change improves trust, task completion, or user-correction behavior. *Example:* Evidence highlighting in the citation UI increases correct user acceptance of agent answers.

The point of the taxonomy is hygiene. A Judge Hypothesis cannot be tested with an A/B test on production users. A Risk Hypothesis cannot be validated with a satisfaction survey. A model-quality hypothesis is not the same as a judge-quality hypothesis. Naming the type prevents the wrong experimental design.

The Multimodal Experimentation Stack

Experimentation sits inside a layered operating model. Each layer answers a different question.

Layer	Question	Output
1. Problem / Risk Language	What harm or product gap motivates the experiment?	Named risk or failure pattern
2. Hypothesis	What do we believe will improve?	Testable belief (CLAIM)
3. Intervention	What exactly changes?	V1 / V2 diff
4. Experimental Design	How will we compare fairly?	Paired, A/B, shadow, canary, ablation, perturbation
5. Measurement	Which contracts, metrics, judges, and slices?	Evaluation plan
6. Statistical Discipline	How much confidence do we need?	Sample size, CIs, paired tests

7. Guardrails	What must not regress?	Safety and quality constraints
8. Decision	What action follows?	Ship, ramp, hold, rollback, investigate
9. Learning Archive	What did we learn?	Reusable organizational knowledge

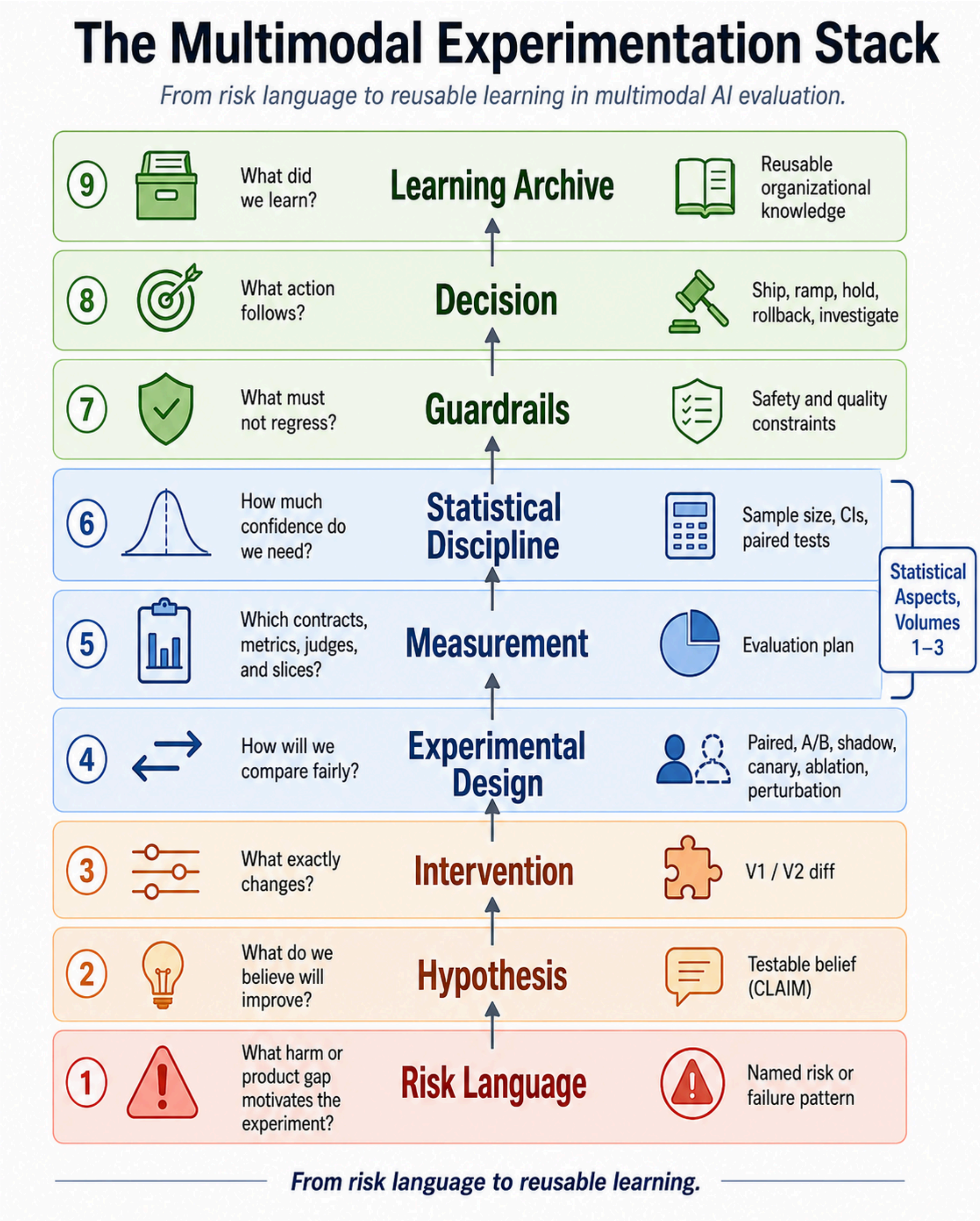
This stack is the experimentation counterpart to the Statistical Aspects trilogy.

Volume 1 asks whether the evidence is being **read** responsibly.

Volume 2 asks whether the evidence was **produced** defensibly.

Volume 3 asks whether the organization makes that production **repeatable**.

Hypothesis-driven experimentation asks whether the organization is actually **learning** from any of it.



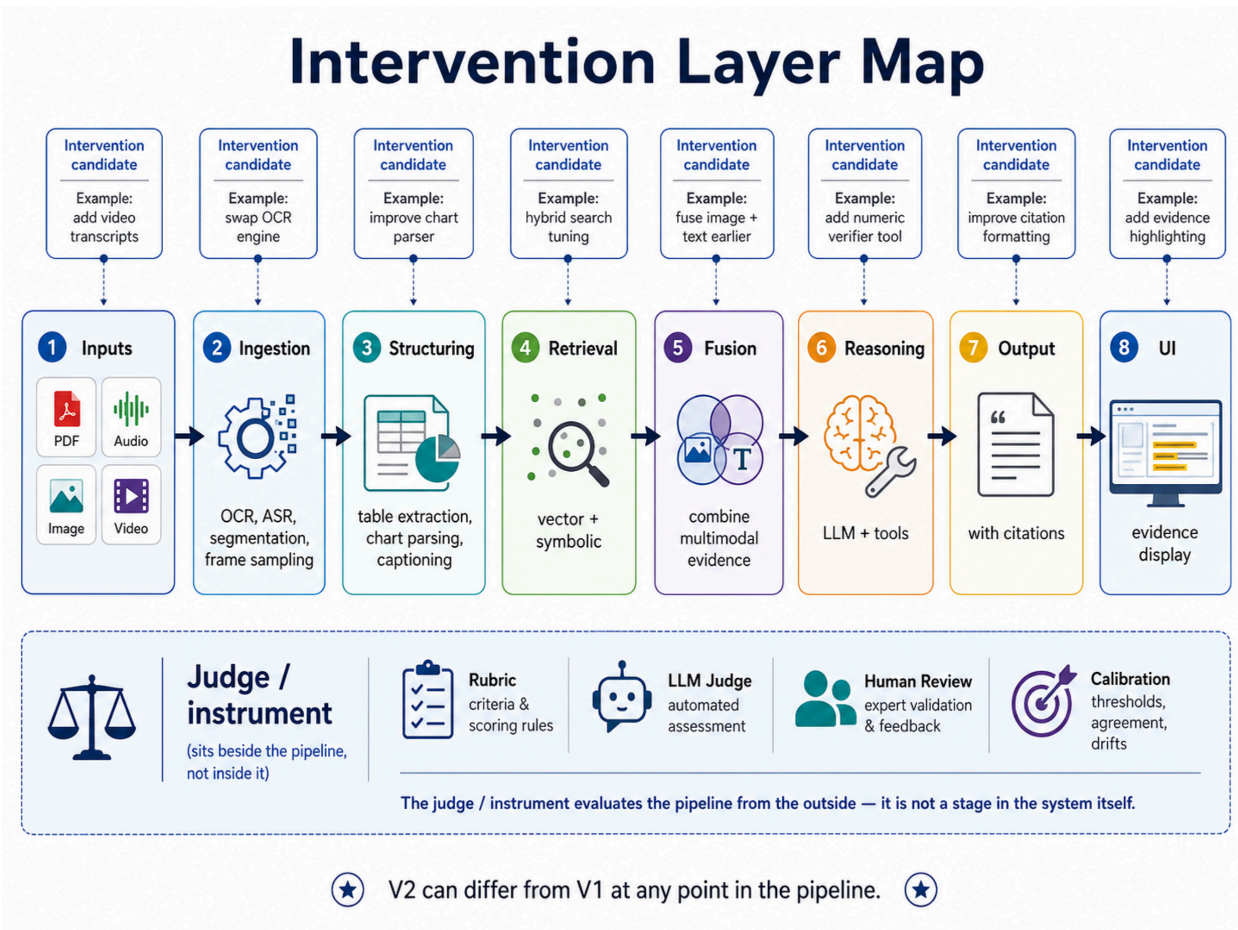
The Multimodal Experimentation Stack
From risk language to reusable learning in multimodal AI evaluation.
9
What did we learn?
Learning Archive
Reusable organizational knowledge
8
What action follows?
Decision
Ship, ramp, hold, rollback, investigate
7
What must not regress?
Guardrails
Safety and quality constraints
6
How much confidence do we need?
Statistical Discipline
Sample size, CIs, paired tests
5
Which contracts, metrics, judges, and slices?
Measurement
Evaluation plan
Statistical Aspects, Volumes 1–3
4
How will we compare fairly?
Experimental Design
Paired, A/B, shadow, canary, ablation, perturbation
3
What exactly changes?
Intervention
V1 / V2 diff
2
What do we believe will improve?
Hypothesis
Testable belief (CLAIM)
1
What harm or product gap motivates the experiment?
Risk Language
Named risk or failure pattern
From risk language to reusable learning.

Why Multimodal AI Experiments Are Harder Than Text-Only AI Experiments

Multimodal AI systems multiply the surface area of experimentation. The "treatment" can live at almost any layer of the pipeline.

Possible intervention points include image preprocessing, OCR, ASR, audio denoising, speaker diarization, video frame sampling, document parsing, page segmentation, table extraction, chart parsing, caption generation, retrieval, reranking, modality fusion, prompting, tool orchestration, memory policy, safety policy, judge choice, and UI evidence display.

Intervention Layer Map



That long list creates five structural complications.

Interventions happen at many layers

V2 may differ from V1 because of parser changes, retriever changes, preprocessing changes, judge changes, UI changes, or orchestration changes — without the LLM itself changing at all. Every experiment must specify *where in the pipeline* the intervention occurs.

Effects appear downstream

A better OCR pipeline may improve Factuality three steps later. A chart parser may improve Cross-Modal Consistency only in chart-heavy documents. A citation UI may improve user trust without any model change at all. Local component metrics will mislead you if you do not measure downstream outcomes.

Averages hide slice-level harm

A variant that improves the headline number may quietly damage multilingual documents, accented speakers, dense tables, medical charts, low-light images, long-form video, or domain-specific diagrams. Slice-level reporting is not optional in Multimodal evaluation. It is the only way to detect the harm averages conceal.

Guardrails matter more

A change can improve one Quality Contract while damaging another. Aggressive OCR cleanup may improve text readability but delete figure captions, collapsing Cross-Modal Consistency. A stricter safety filter may reduce attack success rate while breaking legitimate workflows. Without explicit guardrails, the team trades one harm for another and calls it progress.

Judges are part of the experiment

The judge — whether a rubric, an LLM, a human panel, or a hybrid — is itself a measurement instrument. Judge model version, judge prompt, calibration date, inter-rater agreement, and drift status must all be fixed or reported as part of the experimental design.

If the judge changes, the experiment changes.

An uncalibrated judge is not a measurement instrument — it is noise wearing the costume of evidence. This is why the Statistical Aspects work treats judge calibration, inter-rater agreement, and drift monitoring as core practice, not optional rigor.

Hypotheses for Agentic AI Systems

Multimodal AI is increasingly becoming agentic. The system does not just see, hear, and read — it also plans, calls tools, retrieves memory, reflects, and replans. That shift multiplies the hypothesis surface in three ways.

First, interventions can live at every step of the AI agent's trajectory. A change might happen in the planner ("decompose differently"), the router ("call the chart-extraction tool before invoking the LLM"), the tool catalog ("add a numeric-verification tool"), the memory policy ("evict older retrieval results differently"), the reflection step ("self-critique before final answer"), or the termination condition ("stop after N steps or sufficient evidence"). Each is a candidate for hypothesis-driven testing. Each requires its own measurement plan.

Second, new hypothesis shapes emerge that do not exist in single-shot systems:

- **Trajectory Hypothesis** — A change improves the *path*, not just the endpoint. *Example:* Adding a self-critique step reduces total tool calls without lowering answer quality.
- **Tool-Selection Hypothesis** — A change improves the policy that decides which tool to call. *Example:* The router selects the chart parser before the LLM when chart language appears in the query.
- **Multi-Turn Consistency Hypothesis** — A change improves coherence across turns. *Example:* A persistent grounding cache reduces contradiction rate on follow-up questions.
- **Failure-Recovery Hypothesis** — A change improves what the agent does when a step fails. *Example:* Fallback retrieval reduces null-answer rate on missing-document queries.

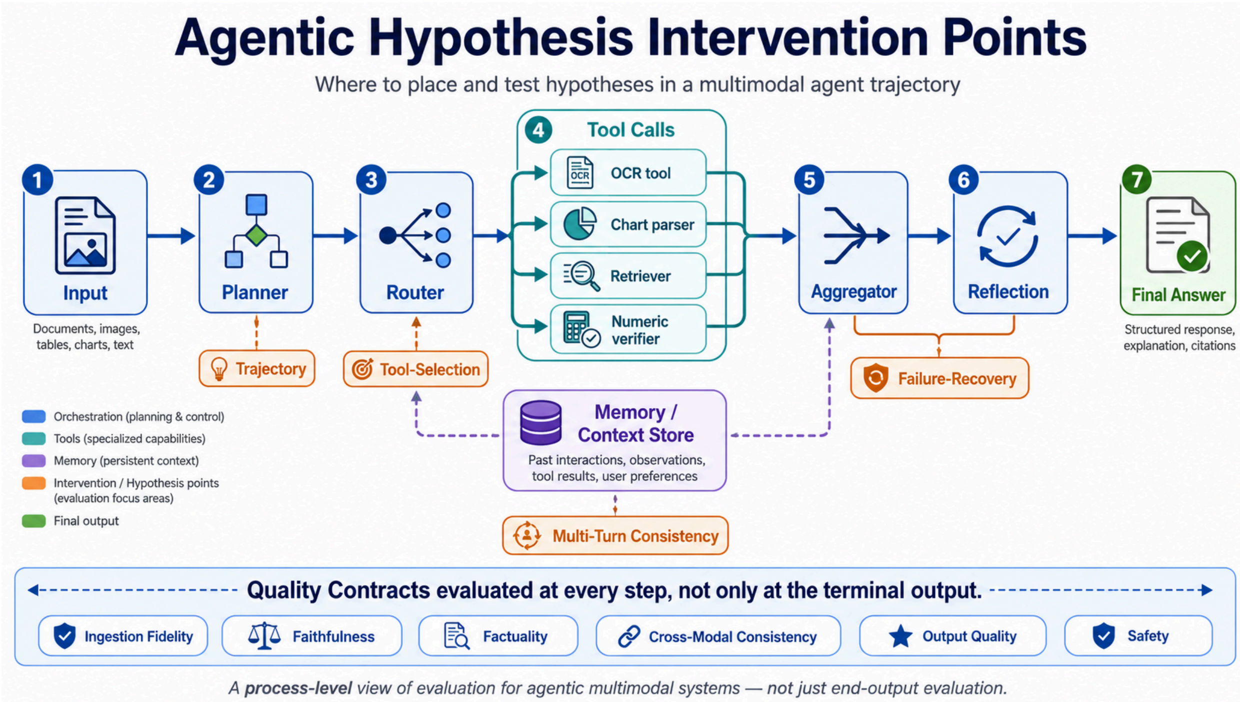
Third, Quality Contracts must be evaluated trajectory-wide, not only on the final answer. Faithfulness can fail mid-trajectory even when the final answer happens to be grounded. Cross-Modal Consistency can collapse on intermediate tool outputs that downstream reasoning then papers over. Safety can fail in a single tool call even when the final response is benign.

A useful frame:

In a single-shot system, the hypothesis tests an output. In an Agentic AI system, the hypothesis tests a process.

The implication is operational. Agentic AI experiments need step-level evaluation hooks, replayable trajectories, and trajectory-diff tools. The pre-registration must specify not only "what does V2 do differently" but "where in the trajectory does the difference appear, and what downstream effects do we expect?" Without that, an agent-level experiment is just a black-box A/B with extra latency — measurement without diagnosis.

Agentic Hypothesis Intervention Points



Experiment Types for Multimodal AI Evaluations

Experimentation is broader than A/B testing.

A/B testing is one experiment type. It is not the whole discipline.

Eight experiment types belong in a Multimodal team's toolkit.

1. Offline paired evaluation. Same cases scored under V1 and V2. Best for model, prompt, parser, OCR/ASR, retrieval, tool-policy, and judge comparisons. Pairing dramatically reduces noise because each case is its own control.

2. Online A/B testing. Real users routed to V1 or V2. Best for user satisfaction, trust, task completion, retention, correction rate, and human override rate. Not appropriate when V2's failure modes carry safety or compliance risk, and not appropriate without an explicit ethics and consent posture in regulated domains.

3. Shadow testing. V2 runs silently alongside V1; users see only V1's output. Best for high-risk settings — healthcare, finance, legal, industrial, safety-critical agentic systems — where exposing users to V2 is premature.

4. Canary testing. V2 rolls out to a small, controlled slice of production traffic. Best for catching latency regressions, drift, and real-world distribution effects before full ramp.

5. Ablation experiments. Remove or isolate one component to measure its contribution. Best for understanding the value of chart parsers, retrievers, memory layers, tools, or fusion strategies.

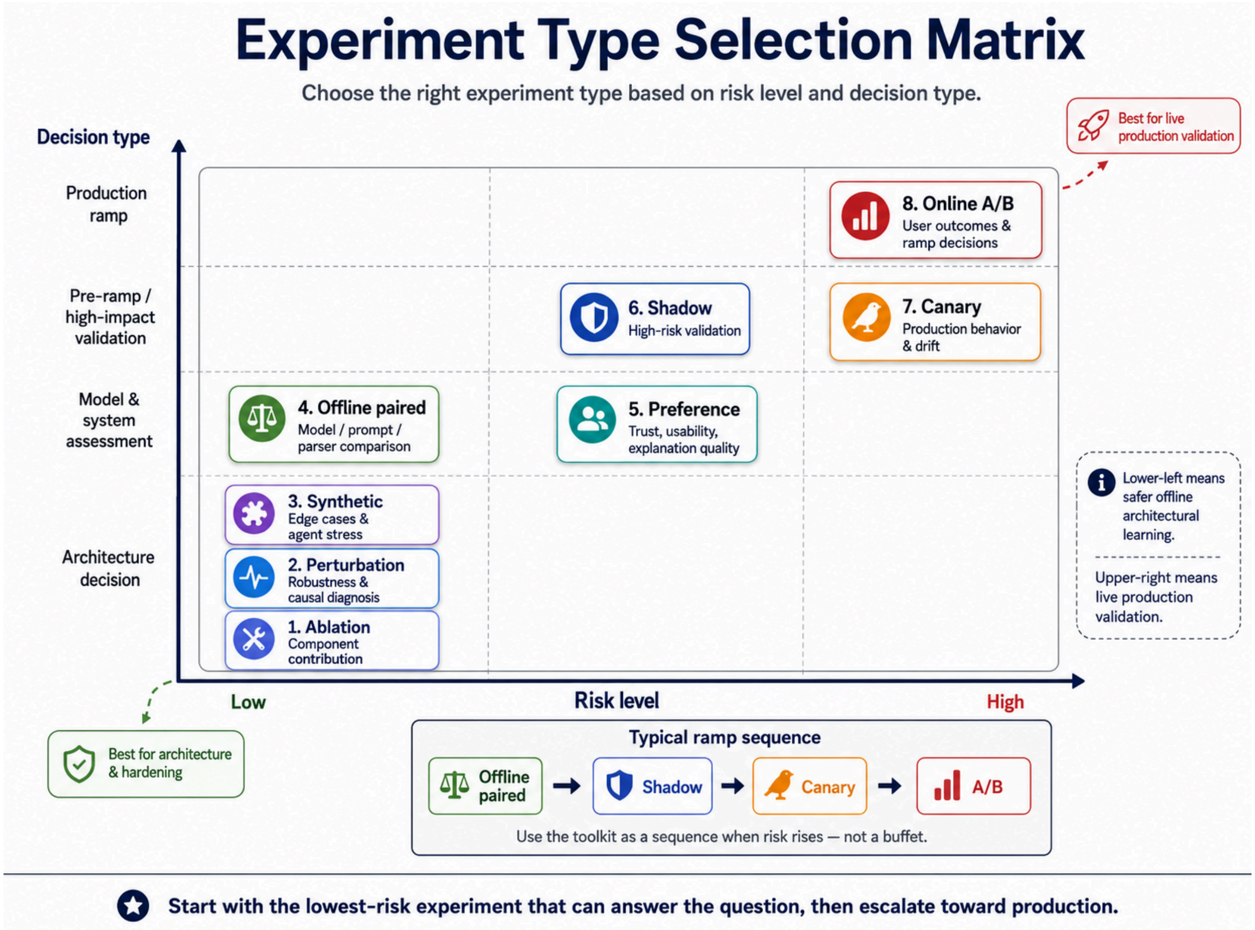
6. Counterfactual / perturbation experiments. Modify one part of the Multimodal input — blur an image, remove captions, add audio noise, swap chart labels, rotate a scan, remove table headers, change input language, alter visual evidence while keeping text constant — and observe behavior. Best for robustness and causal diagnosis.

7. Human preference experiments. Users or domain experts compare paired outputs. Best for explanation quality, groundedness, trust, usability, tone, and perceived helpfulness.

8. Synthetic persona / scenario experiments. Synthetic users, documents, images, audio, or workflows. Best for edge cases, rare events, agentic workflow stress testing, and early product learning before real data is available.

Experiment Type	Best Used For	Decision Supported	Risk Level
Offline paired	Model / prompt / parser / retrieval comparison	Pre-ramp decisions	Low
Online A/B	Trust, satisfaction, task completion	Ship / ramp	Medium
Shadow	High-risk validation	Pre-canary	Low (to user)
Canary	Production behavior, latency, drift	Ramp / hold	Medium
Ablation	Component contribution	Architecture	Low
Perturbation	Robustness, causal diagnosis	Redesign / harden	Low
Preference	Explanation, trust, usability	Ship / iterate	Low
Synthetic	Edge cases, agent stress	Investigate / harden	Low

Experiment Type Selection Matrix



The Experiment Portfolio

A team is not judged by the quality of a single experiment. It is judged by the quality of its **experiment portfolio** — the mix of experiments it runs over time, and how that mix produces durable learning.

Three modes belong in a healthy portfolio.

Confirmatory experiments. Test a pre-registered hypothesis. The team believes V2 will improve a specific Quality Contract for a specific slice by a specific amount. CLAIM, the T-19 brief, and the pre-registration discipline target this mode. Confirmatory experiments are how teams ship.

Exploratory experiments. Probe for unknown failure modes, unexpected slices, and unmodeled risks. There is no pre-registered hypothesis. The team is *looking*, not *testing*. Counterfactual perturbations, synthetic stress scenarios, and adversarial probes belong here. Exploratory experiments are how teams discover what to test next.

Monitoring experiments. Run continuously against production traffic and a frozen calibration set. Judge drift checks, Quality Contract regression alarms, and slice-level production gauges belong here. Monitoring experiments are how teams notice that yesterday's evidence is no longer current.

Mode	Question Answered	Typical Cadence	Typical Cost	Primary Output
Confirmatory	Did the planned change work?	Per release	Medium–High	Decision: ship / ramp / hold / rollback
Exploratory	What might be broken that we are not testing for?	Quarterly or by signal	Low–Medium	New hypotheses for the confirmatory pipeline
Monitoring	Has yesterday's evidence gone stale?	Continuous	Low (amortized)	Drift alerts and re-test triggers

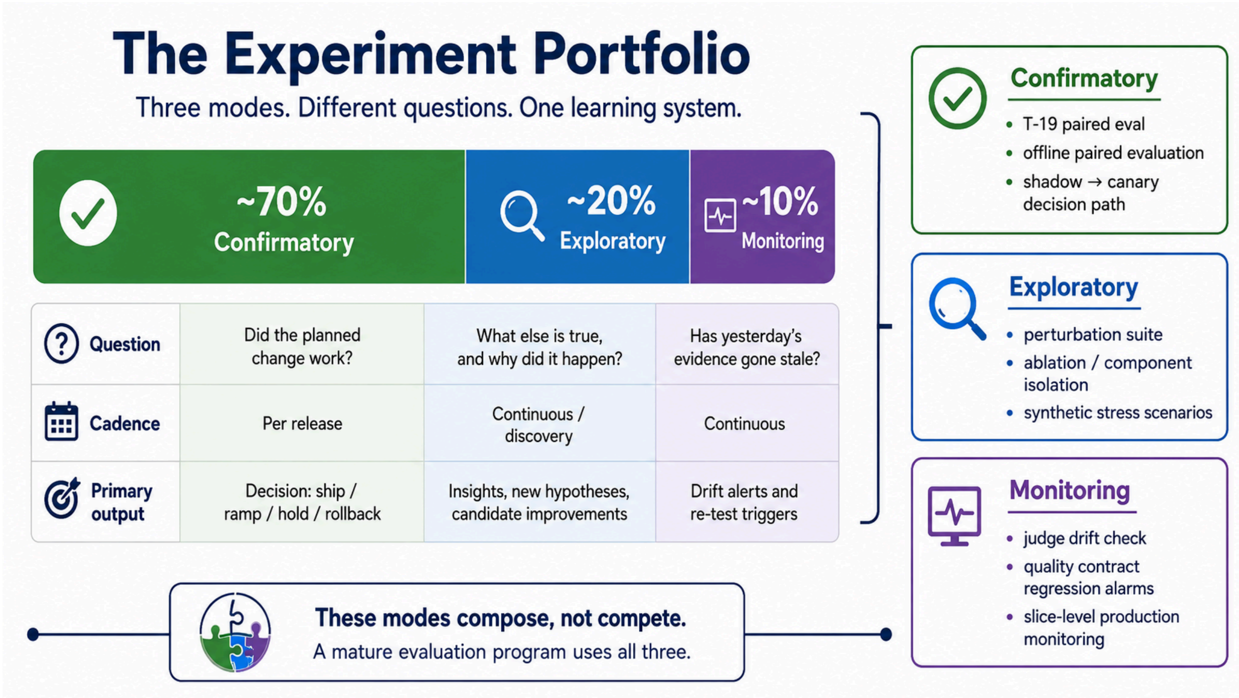
The three modes are not interchangeable. A team running only confirmatory experiments will improve quickly on the dimensions it already knows about, and slowly on the dimensions it does not. A team running only exploratory experiments will accumulate findings without shipping. A team running only monitoring will detect regressions but never propose interventions.

A useful heuristic for a mature Multimodal evaluation program: roughly **70% confirmatory, 20% exploratory, 10% monitoring**. Younger programs lean heavier on exploratory work because they do not yet know what to test. Older programs lean heavier on monitoring because the system surface has stabilized.

The portfolio frame also surfaces an economic question that single-experiment thinking hides: **experimentation has a budget.** Human raters, GPU hours, calibrated judges, and real-user exposure all cost something. A team that spends a 5,000-case paired eval to ship a 0.4-point lift on an already-saturated contract is spending its experimentation budget badly, even if the experiment is methodologically clean. The right question is not "is this experiment well-designed?" but "is this the experiment we should be running right now?"

The portfolio maps onto the AMI Cycle. The three modes correspond almost one-to-one to the Analyze → Measure → Improve loop that anchors the broader Multimodal AI Evaluations framework. **Exploratory experiments** are the Analyze phase: they surface what to test. **Monitoring experiments** are the Measure phase: they keep the instruments honest. **Confirmatory experiments** are the Improve phase: they ship the changes that move quality. Teams that run only one mode are running only one-third of the cycle.

The Experiment Portfolio



The Experimentation Maturity Model

A team can read this document and still not know where to start, because "start where you are" requires knowing where you are. The five-level ladder below is a self-assessment tool — a way to locate a team's current experimentation discipline and identify the next adjacent step.

Level	Status	Behavior
1	**Reactive**	Experiments happen ad hoc, in response to incidents or stakeholder pressure. No CLAIM brief. No pre-registration. The "experiment" is whatever the team did that week. Decisions are made by whoever speaks last in the release meeting.
2	**Pre-Registered**	Teams write down hypotheses, primary metrics, and decision rules before V2 results are seen. T-19 briefs exist for major releases. Confirmatory experiments dominate the portfolio (90%+). Slice plans are inconsistent.
3	**Sliced**	Pre-registration includes explicit slice plans and guardrails on critical slices. Average-versus-slice tradeoffs are surfaced before ramp.

		Judge calibration is monitored. Failure patterns are named and partially prevented.
4	**Compounding**	The Learning Archive is real and consulted. Past experiments inform new hypotheses. The portfolio includes exploratory and monitoring work, not only confirmatory. Right-sizing decisions happen explicitly. Agentic experiments include trajectory-level evidence.
5	**Self-Improving**	New failure modes are turned into pre-registered hypotheses within one sprint of discovery. Judge calibration is automated. The Risk-Aware Operating Loop drives ROI on experimentation budget. *Investigate* decisions reliably produce next-experiment hypotheses, not stalled tickets.

A team has reached Level 5 when an outsider can pick up any closed experiment from the Archive, find its pre-registered hypothesis, see the slice plan, read the decision memo, and understand both *what changed* and *what was learned* — all without asking anyone.

This experimentation ladder is the companion to — not a replacement for — the broader Evaluation Maturity Model in the Multimodal AI Evaluations framework. A team can be at different levels on the two ladders. Sophisticated evaluation discipline does not automatically produce sophisticated experimentation discipline; in some programs the gap is wide and visible. Naming the gap is the start of closing it.

The most common stuck point is between Levels 2 and 3. A team that has learned to pre-register a hypothesis but does not yet pre-register a slice plan will ship variants that improve the headline number while quietly regressing critical segments. The discipline of writing down the slice plan *before* the results arrive is what makes the next level reachable.

Mapping Experiments to the Quality Contracts

The Multimodal AI Evaluations framework defines five Quality Contracts — Ingestion Fidelity, Faithfulness, Factuality, Cross-Modal Consistency, and Output Quality — with Safety as a sixth applied contract. Hypotheses without a named contract tend to produce metrics without a clear interpretation. Hypotheses anchored to a contract produce decisions.

Quality Contract	Experiment Hypothesis	Primary Metric	Guardrails
Ingestion Fidelity	New OCR improves scanned PDF extraction	Extraction accuracy	Caption loss, table distortion

Faithfulness	Grounded retrieval reduces unsupported claims	Faithfulness rate	Latency, refusal quality
Factuality	Domain verifier reduces numerical errors	Factual accuracy	Over-refusal, UX degradation
Cross-Modal Consistency	Chart parser improves chart-text alignment	Cross-modal agreement	Hallucination rate, latency
Output Quality	Structured answer template improves usefulness	Human preference / rubric score	Faithfulness, factuality
Safety	Injection detector reduces unsafe tool actions	Attack success rate	False positives, task completion

The Quality Contracts prevent vague experiments by anchoring hypotheses to measurable dimensions. They also force a team to declare *which* contract a change is supposed to improve, which in turn surfaces the guardrails that might quietly fail.

The Hypothesis-Driven Evaluation Workflow

A nine-step workflow turns the stack into practice. Applied to the running example — table-aware chunking plus chart-structure extraction for scanned financial reports — it looks like this.

Step 1: Start with Risk Language or a product gap. Analysts report that the agent fabricates numbers when summarizing tables and misreads chart axes. The Risk Language is *numerical hallucination in financial reasoning* and *chart-text disagreement*.

Step 2: Translate into Eval Language. Numerical hallucination maps to Factuality. Chart-text disagreement maps to Cross-Modal Consistency. Ingestion Fidelity is the upstream contract that gates both.

Step 3: Write the hypothesis using CLAIM. *Change:* Table-aware chunking + chart-structure extraction. *Logic:* Preserving table structure and parsing chart axes gives the LLM cleaner grounded inputs. *Assessment:* Paired offline eval on 500 scanned reports. *Impact:* ≥4-point Faithfulness lift; ≥5-point Cross-Modal Consistency lift. *Mitigation:* No regression on Ingestion Fidelity, latency, or non-table queries.

Step 4: Choose the experiment type. Offline paired evaluation first, shadow test second, canary third. No A/B until canary is clean.

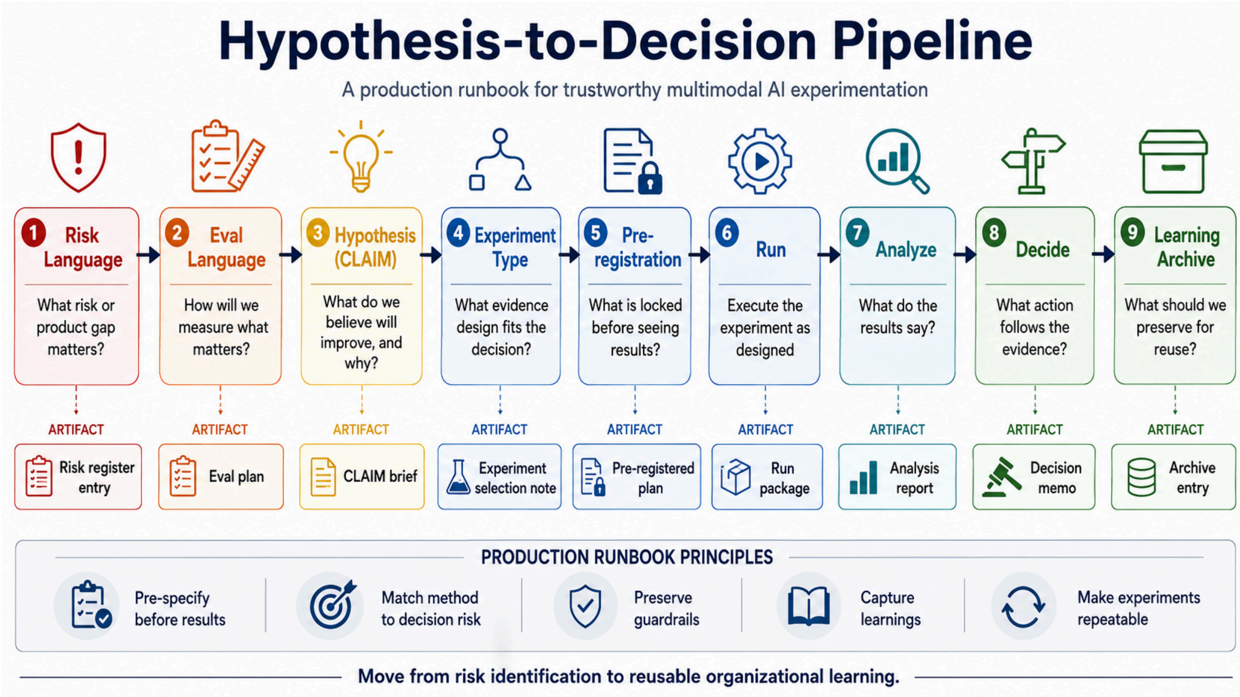
Step 5: Pre-register the experiment. Lock the eval set, the slice plan, the judge version, the primary metric, the guardrails, and the decision rule *before* any V2 results are seen.

Step 6: Run the experiment. Protect holdout integrity. Use identical cases for V1 and V2. Record the world state — model versions, retriever index date, judge model and prompt version, hardware, queue load.

Step 7: Analyze the evidence. Report effect size with confidence intervals. Run paired tests. Break down by slice — scanned vs. born-digital, dense-table vs. chart-heavy, English vs. multilingual. Verify the judge has not drifted during the experiment.

Step 8: Decide. Apply the pre-registered decision rule. Ship, ramp, hold, rollback, investigate, redesign, or update the eval set.

Step 9: Archive the learning. File the experiment in the Learning Archive: what was tested, what was found, which slices moved, which guardrails were stressed, and what new hypotheses emerged.



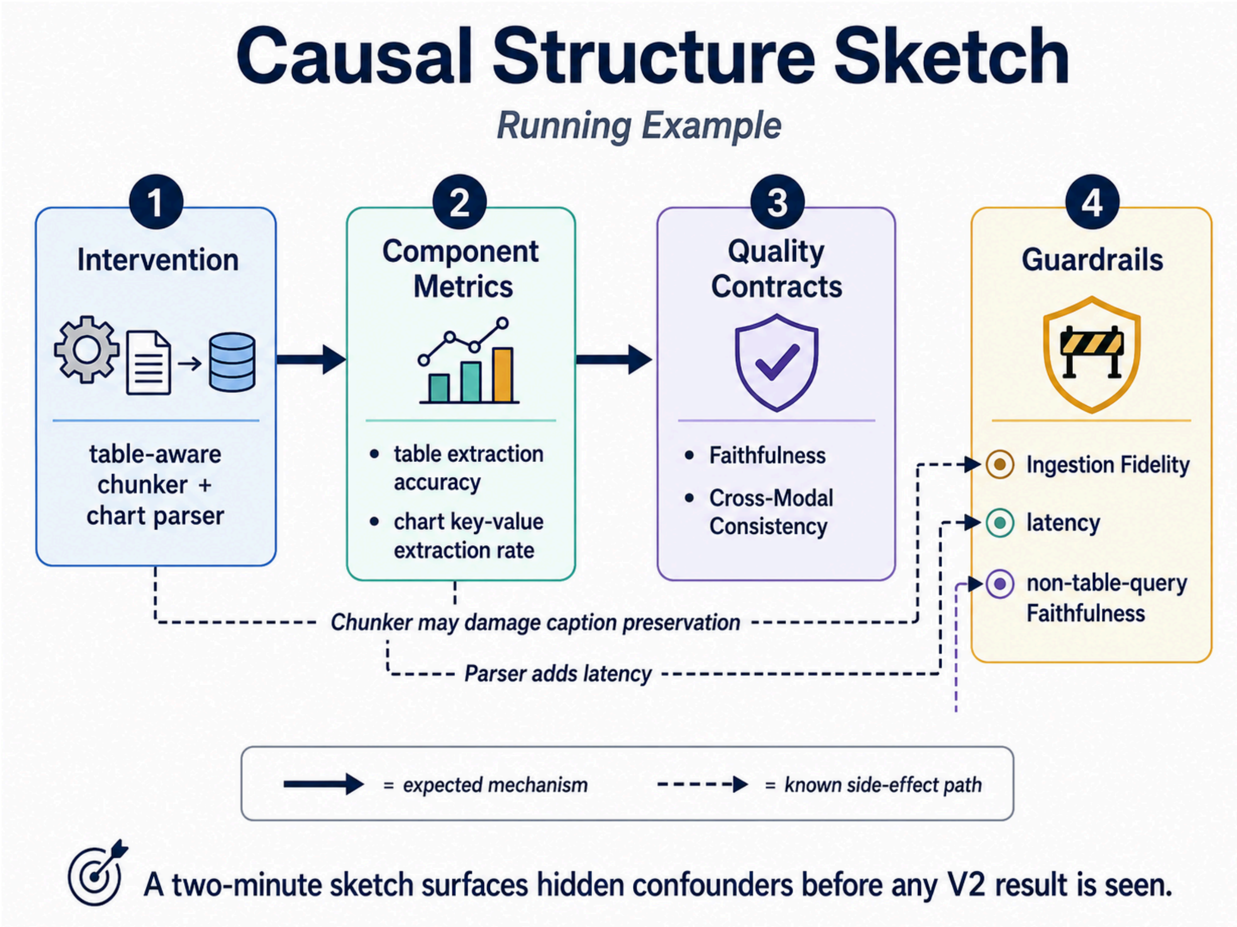
Pre-Registration and Statistical Discipline

Pre-registration is not bureaucracy. It is how teams prevent themselves from turning post-hoc discoveries into fake proof.

Before any V2 result is observed, the team writes down the hypothesis, the intervention, the control and treatment, the primary Quality Contract, the minimum meaningful effect, the guardrail metrics, the sample-size and power logic, the slice plan, the judge or instrument and its calibration status, the stopping rule, the decision rule, and the world-state assumptions (model versions, index dates, traffic patterns).

That document is the experiment's pre-registration. It exists so that when the results arrive, the team cannot retroactively rewrite the question to match the answer. In the Volume 3 operating model, the pre-registration is what gets entered into the Quality Contract registry; any deviation between the pre-registration and the actual run produces an entry in the Statistical Debt Register that must be paid down before another experiment touches the same contract.

A small discipline that pays off out of proportion to its cost: **before pre-registering, sketch the causal structure.** A two-minute diagram on a whiteboard — *Intervention* → *Component metric* → *Quality Contract* → *Guardrail*, with arrows showing the expected mechanism and the side effects — surfaces hidden confounders, downstream effects, and missing measurements that a prose hypothesis often hides. Most "we don't know what moved the score" post-mortems trace back to a causal graph the team never drew.



Statistical discipline does the rest of the work:

- **Sample size** ensures the experiment can actually detect the effect the team cares about.
- **Confidence intervals** show how much uncertainty surrounds the point estimate.
- **Paired comparisons** strip out case-level variance and reveal smaller true effects.
- **Stratification** exposes slice-level harm that the average would hide.
- **Judge calibration** protects measurement validity against drift and bias.
- **Multiple-comparison correction** prevents cherry-picking across many slices.
- **Holdout integrity** protects generalization — a leaked eval set is no longer an eval set.
- **Replication** protects against one-off results that will not survive the next data refresh.

A necessary caveat:

Hypothesis-driven evaluation improves causal learning, but it does not magically prove causality. The strength of the causal claim depends on the experimental design, isolation of the intervention, randomization, control of confounders, and stability of the measurement instrument.

Pre-registration plus paired design plus calibrated judges plus slice reporting does not guarantee truth. It guarantees that when the team is wrong, they will know it earlier and cheaper.

Decision Rules: Ship, Ramp, Hold, Rollback, Learn

Every experiment must end in a decision. The decision should be one of:

- **Ship.** Evidence clears the threshold, guardrails pass, slices are clean.
- **Ramp.** Evidence is positive but production validation is still needed (canary, shadow → A/B).
- **Hold.** Evidence is insufficient; do not regress, do not advance.
- **Rollback.** A production guardrail has been breached; revert.
- **Investigate.** Effect is unclear or slices disagree; design a follow-up experiment.
- **Redesign.** The hypothesis was disproven; rethink the intervention.
- **Update eval set.** The experiment revealed a measurement gap; the eval set itself needs work before another experiment is meaningful.

A good Multimodal experiment does not end with a p-value. It ends with a decision and a learning artifact.

For the running example, the offline paired evaluation shows a Faithfulness lift of +5.2 points (95% CI: +3.8 to +6.6) and a Cross-Modal Consistency lift of +6.1 points, but the dense-table slice shows a 1.9-point Ingestion Fidelity regression — caption text is being absorbed into table cells. The pre-registered decision rule blocks ramp. The decision is *Investigate*. A targeted fix to caption preservation becomes the next experiment.

That outcome is not a failure. It is the system working as designed.

What would have shipped without the experiment. If the team had relied on the headline dashboard alone, V2 would have shipped to production on the strength of the +5.2 Faithfulness lift and the +6.1 Cross-Modal Consistency lift. The dense-table regression would have surfaced weeks later — in analyst escalations about specific report families, in compliance inquiries about misquoted captions, or as a slow erosion of trust that nobody on the team could explain. The experiment did not just produce a decision. It bought the team the right to be surprised in evaluation rather than in production.

Failure Patterns to Avoid

Thirteen failure patterns recur across Multimodal AI evaluation programs. Each has a prevention.

1. The Dashboard Masquerading as an Experiment. "V2 is better" with no hypothesis, no pre-registration, no slice plan, no guardrails. *Prevention:* Require a CLAIM brief before any release-decision conversation.

2. The Intervention Was Not Isolated. Model, prompt, retriever, parser, and judge all changed in the same release. Nobody knows what caused the lift. *Prevention:* One change per experiment, or explicit ablations.

3. The Average Improved, the Critical Slice Regressed. Overall score moves up; multilingual documents, accented audio, dense tables, or safety-critical cases silently get worse. *Prevention:* Slice plan in the pre-registration; guardrails on critical slices.

4. The Judge Changed During the Experiment. The judge model provider silently updated. The experiment measured judge drift, not system improvement. *Prevention:* Pin judge version. Monitor judge agreement against a stable human-rated calibration set. Re-run if drift is detected.

5. The Winning Metric Was Picked After Seeing Results. Classic HARKing — hypothesizing after the result is known. The team tests many metrics and celebrates the one that moved. *Prevention:* Lock the primary metric in pre-registration. Other metrics become exploratory, not confirmatory.

6. Online A/B Test Used Where Shadow Test Was Needed. Real users were exposed to unproven or unsafe behavior because the team skipped a step. *Prevention:* Risk-tier the experiment first. High-risk changes go through shadow before users see them.

7. No Learning Archive. The experiment was discussed once in Slack, never written down, and forgotten by the next quarter. *Prevention:* No experiment is closed until its Learning Archive entry is filed.

8. Local Component Metric Improved, Downstream Contract Did Not. Component-level tests showed the new OCR engine extracted text four points better. Downstream Faithfulness and Cross-Modal Consistency did not move — or moved the wrong way. *Prevention:* Treat local-metric improvements as leading indicators, never as decisions. Every component-level hypothesis must include a downstream Quality Contract measurement.

9. The Eval Set Co-Evolved with the System. The team kept tuning the eval set as the system improved. The score went up, but it was measuring a progressively easier test. *Prevention:* Maintain a frozen "core" holdout that does not change between versions. Eval-set additions require their own change log, pre-registration, and version stamp.

10. The Cost of Evidence Exceeded the Value of the Decision. The team ran a 5,000-case paired eval with three human raters and a calibrated LLM judge to ship a 0.4-point improvement on a contract that was already well above target. Months of measurement work, weeks of human grading, marginal product impact. *Prevention:* Right-size the experiment. Match measurement cost to the importance of the decision. Not every change deserves a 500-case paired eval, and some changes deserve more than one.

11. The Team Optimized the Eval, Not the Outcome. Iterations made the eval score climb, but downstream user trust, completion rate, and override behavior did not move. The eval became the goal instead of the proxy. *Prevention:* Anchor at least one online, behavioral, or human-preference metric to every Quality Contract. If the eval improves but the user does not, the eval has drifted from the goal — and the eval, not the system, is what needs to change.

12. Trajectory Hooks Not Wired Before the Agentic Experiment. The team ran an agent-level A/B but had no instrumentation to grade intermediate steps. V2 won at the terminal output, but nobody can tell whether it won by reasoning better or by getting luckier on retrieval. *Prevention:* For any agentic intervention, instrument trajectory diff and step-level Quality Contract evaluation *before* enrolling cases. An agent-level A/B without step-level evidence is a black-box A/B with extra latency.

13. Agent State Was Not Deterministic Across V1 and V2. V1 and V2 used different memory states, retrieval index dates, or tool versions during the paired comparison. The comparison was unfair, the lift was an artifact of world-state, not the intervention. *Prevention:* Lock the world-state before the experiment: same retrieval index date, same tool versions, same memory policy seed. Document it in the pre-registration's *world-state assumptions* field.

The Hypothesis-Driven Multimodal Evaluation Brief

The artifact that makes all of this real is the **T-19 Hypothesis-Driven Multimodal Evaluation Brief** — a one-page (or one-document) template that bridges product intent, evaluation design, statistical discipline, and decision-making.

T-19 — Hypothesis-Driven Multimodal Evaluation Brief

1. Experiment name.
2. Risk Language / product gap.
3. Hypothesis (CLAIM format).
4. Intervention.
5. Control and treatment.
6. Quality Contract targeted.
7. Primary metric.
8. Guardrail metrics.

9. Modalities involved.
10. Target slices / strata.
11. Experiment type.
12. Sample size / power logic.
13. Judge / instrument and calibration status.
14. Pre-registered decision rule.
15. Rollout plan.
16. Learning Archive entry.

T-19 — Worked Example

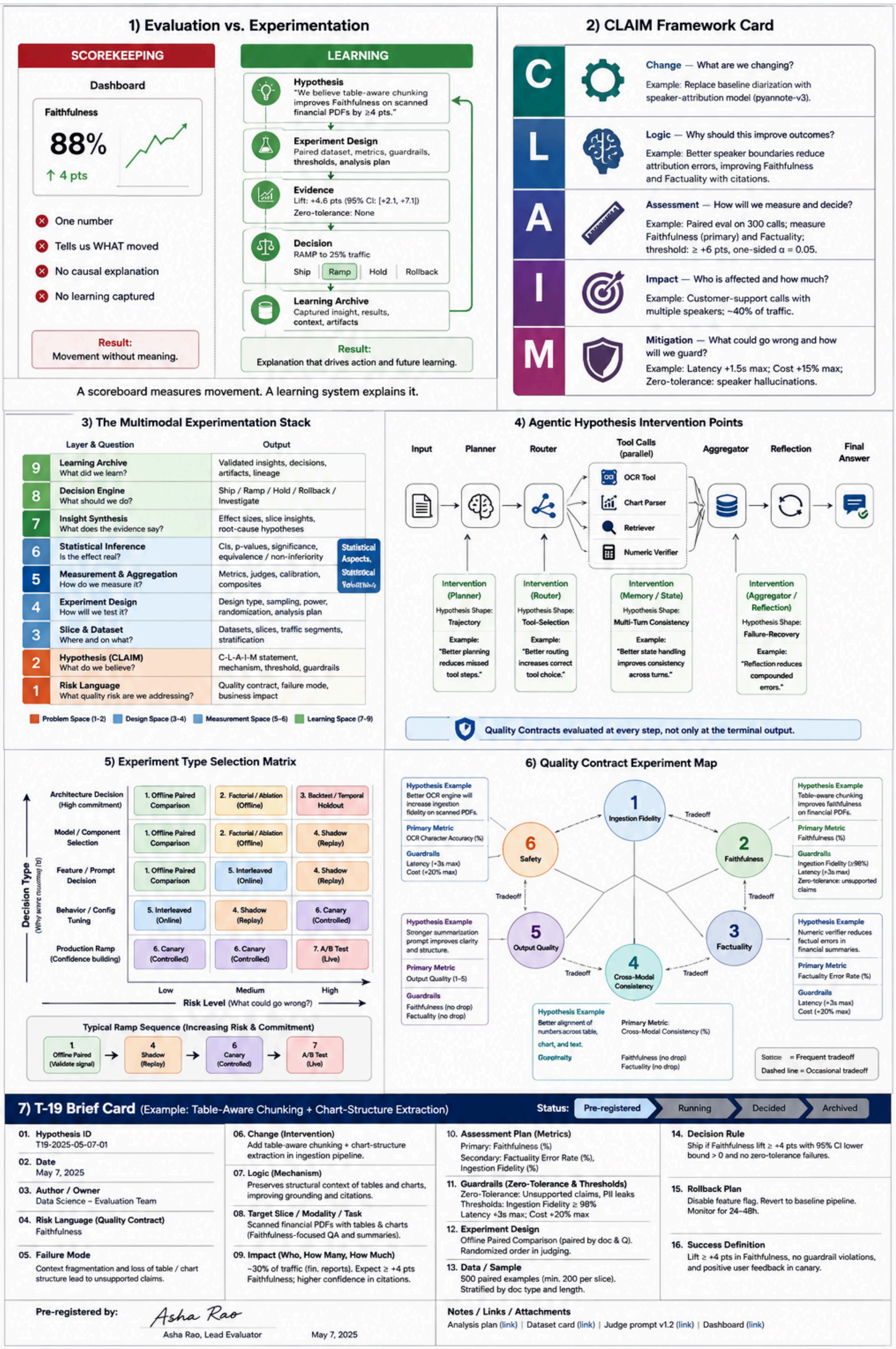
Populated end-to-end for the running example: table-aware chunking + chart-structure extraction.

#	Field	Value
1	Experiment name	E-2026-Q2-FIN-INGEST-01
2	Risk Language / product gap	Numerical hallucination in financial summaries; chart-text disagreement on scanned reports
3	Hypothesis (CLAIM)	Adding table-aware chunking and chart-structure extraction will improve Faithfulness (≥4 pts) and Cross-Modal Consistency (≥5 pts) on scanned financial PDFs, without regressing Ingestion Fidelity, latency, or non-table queries
4	Intervention	Replace flat token chunker with table-aware chunker; insert chart parser between page segmentation and retrieval indexing
5	Control / treatment	V1 = current pipeline (flat chunker, no chart parser); V2 = V1 + table-aware chunker + chart parser
6	Quality Contract targeted	Primary: Faithfulness, Cross-Modal Consistency. Upstream: Ingestion Fidelity
7	Primary metric	Paired Faithfulness rate, judged by Judge-FIN-v3 against analyst-rated gold answers
8	Guardrail metrics	Ingestion Fidelity ≥ 98%; end-to-end latency ≤ 3.0s p95; Factuality on numerical summaries (no regression); non-table-query Faithfulness (no regression)
9	Modalities involved	Scanned PDFs (image), embedded tables, embedded charts, OCR-derived text

10	Target slices / strata	Scanned vs. born-digital; dense-table vs. chart-heavy; English-only vs. multilingual
11	Experiment type	Offline paired evaluation → shadow → 10% canary → A/B
12	Sample size / power logic	n=500 paired cases; powered to detect ≥3-pt effect at α=0.05 with paired bootstrap CI
13	Judge / instrument and calibration status	Judge-FIN-v3 (pinned); κ=0.78 against human raters as of 2026-04-15; mid-run drift check scheduled
14	Pre-registered decision rule	Ramp to canary if primary metric clears threshold AND all guardrails pass on every slice; otherwise Investigate
15	Rollout plan	Offline pass → shadow (2 weeks) → 10% canary (1 week) → 50% A/B (2 weeks) → full ship
16	Learning Archive entry	Filed within 5 business days of decision, regardless of outcome

Why n=500? The 500-case sample size was chosen to detect a paired effect of ≥3 percentage points on Faithfulness at α=0.05 with paired bootstrap confidence intervals, assuming case-level standard deviation σ ≈ 0.18 observed in prior runs. The five primary slices (scanned, born-digital, dense-table, chart-heavy, multilingual) each receive roughly 100 cases. Slices below ~80 cases are pre-declared as underpowered: positive movement on those slices is reported as suggestive, not confirmatory, and triggers a follow-up experiment rather than a ramp decision. The pre-registration documents both the headline power calculation and the slice-level power gaps, so the team is honest with itself about what the experiment can and cannot prove.

A team that completes this brief before every experiment will not always be right. But it will be wrong in *legible* ways, which is the precondition for learning. The brief belongs as a versioned artifact in the team's EvalOps repository, not as a wiki page that goes stale.


1) Evaluation vs. Experimentation
SCOREKEEPING
Dashboard
Faithfulness
88%
↑ 4 pts
One number
Tells us WHAT moved
No causal explanation
No learning captured
Result:
Movement without meaning.
LEARNING
Hypothesis
"We believe table-aware chunking improves Faithfulness on scanned financial PDFs by ≥4 pts."
Experiment Design
Paired dataset, metrics, guardrails, thresholds, analysis plan
Evidence
Lift: +4.6 pts (95% CI: [+2.1, +7.1])
Zero-tolerance: None
Decision
RAMP to 25% traffic
Ship
Ramp
Hold
Rollback
Learning Archive
Captured insight, results, context, artifacts
Result:
Explanation that drives action and future learning.
A scoreboard measures movement. A learning system explains it.
2) CLAIM Framework Card
C
Change — What are we changing?
Example: Replace baseline diarization with speaker-attribution model (pyannote-v3).
L
Logic — Why should this improve outcomes?
Example: Better speaker boundaries reduce attribution errors, improving Faithfulness and Factuality with citations.
A
Assessment — How will we measure and decide?
Example: Paired eval on 300 calls; measure Faithfulness (primary) and Factuality; threshold: ≥ +6 pts, one-sided α = 0.05.
I
Impact — Who is affected and how much?
Example: Customer-support calls with multiple speakers; ~40% of traffic.
M
Mitigation — What could go wrong and how will we guard?
Example: Latency +1.5s max; Cost +15% max; Zero-tolerance: speaker hallucinations.
3) The Multimodal Experimentation Stack
Layer & Question
Output
9 Learning Archive
What did we learn?
Validated insights, decisions, artifacts, lineage
8 Decision Engine
What should we do?
Ship / Ramp / Hold / Rollback / Investigate
7 Insight Synthesis
What does the evidence say?
Effect sizes, slice insights, root-cause hypotheses
6 Statistical Inference
Is the effect real?
CIs, p-values, significance, equivalence / non-inferiority
5 Measurement & Aggregation
How do we measure it?
Metrics, judges, calibration, composites
Statistical Aspects,
4 Experiment Design
How will we test it?
Design type, sampling, power, randomization, analysis plan
3 Slice & Dataset
Where and on what?
Datasets, slices, traffic segments, stratification
2 Hypothesis (CLAIM)
What do we believe?
C-L-A-I-M statement, mechanism, threshold, guardrails
1 Risk Language
What quality risk are we addressing?
Quality contract, failure mode, business impact
Problem Space (1-2)
Design Space (3-4)
Measurement Space (5-6)
Learning Space (7-9)
4) Agentic Hypothesis Intervention Points
Input
Planner
Router
Tool Calls (parallel)
OCR Tool
Chart Parser
Retriever
Numeric Verifier
Aggregator
Reflection
Final Answer
Intervention (Planner)
Hypothesis Shape:
Trajectory
Example:
"Better planning reduces missed tool steps."
Intervention (Router)
Hypothesis Shape:
Tool-Selection
Example:
"Better routing increases correct tool choice."
Intervention (Memory / State)
Hypothesis Shape:
Multi-Turn Consistency
Example:
"Better state handling improves consistency across turns."
Intervention (Aggregator / Reflection)
Hypothesis Shape:
Failure-Recovery
Example:
"Reflection reduces compounded errors."
Quality Contracts evaluated at every step, not only at the terminal output.
5) Experiment Type Selection Matrix
Decision Type
Architecture Decision (High commitment)
1. Offline Paired Comparison
2. Factorial / Ablation (Offline)
3. Backtest / Temporal Holdout
Model / Component Selection
1. Offline Paired Comparison
2. Factorial / Ablation (Offline)
4. Shadow (Replay)
Feature / Prompt Decision
1. Offline Paired Comparison
5. Interleaved (Online)
4. Shadow (Replay)
Behavior / Config Tuning
5. Interleaved (Online)
4. Shadow (Replay)
6. Canary (Controlled)
Production Ramp (Confidence building)
6. Canary (Controlled)
6. Canary (Controlled)
7. A/B Test (Live)
Low
Medium
High
Risk Level (What could go wrong?)
Typical Ramp Sequence (Increasing Risk & Commitment)
1 Offline Paired (Validate signal)
4 Shadow (Replay)
6 Canary (Controlled)
7 A/B Test (Live)
6) Quality Contract Experiment Map
1 Ingestion Fidelity
2 Faithfulness
3 Factuality
4 Cross-Modal Consistency
5 Output Quality
6 Safety
Tradeoff
Tradeoff
Tradeoff
Tradeoff
Tradeoff
Hypothesis Example
Better OCR engine will increase ingestion fidelity on scanned PDFs.
Primary Metric
OCR Character Accuracy (%)
Guardrails
Latency (+3s max)
Cost (+20% max)
Hypothesis Example
Table-aware chunking improves faithfulness on financial PDFs.
Primary Metric
Faithfulness (%)
Guardrails
Ingestion Fidelity (≥98%)
Latency (+3s max)
Zero-tolerance: unsupported claims
Hypothesis Example
Stronger summarization prompt improves clarity and structure.
Primary Metric
Output Quality (1–5)
Guardrails
Faithfulness (no drop)
Factuality (no drop)
Hypothesis Example
Numeric verifier reduces factual errors in financial summaries.
Primary Metric
Factuality Error Rate (%)
Guardrails
Latency (+3s max)
Cost (+20% max)
Hypothesis Example
Better alignment of numbers across table, chart, and text.
Primary Metric:
Cross-Modal Consistency (%)
Faithfulness (no drop)
Factuality (no drop)
= Frequent tradeoff
Dashed line = Occasional tradeoff
7) T-19 Brief Card (Example: Table-Aware Chunking + Chart-Structure Extraction)
Status: Pre-registered Running Decided Archived
01. Hypothesis ID
T19-2025-05-07-01
02. Date
May 7, 2025
03. Author / Owner
Data Science – Evaluation Team
04. Risk Language (Quality Contract)
Faithfulness
05. Failure Mode
Context fragmentation and loss of table / chart structure lead to unsupported claims.
06. Change (Intervention)
Add table-aware chunking + chart-structure extraction in ingestion pipeline.
07. Logic (Mechanism)
Preserves structural context of tables and charts, improving grounding and citations.
08. Target Slice / Modality / Task
Scanned financial PDFs with tables & charts (Faithfulness-focused QA and summaries).
09. Impact (Who, How Many, How Much)
~30% of traffic (fin. reports). Expect ≥ +4 pts Faithfulness; higher confidence in citations.
10. Assessment Plan (Metrics)
Primary: Faithfulness (%)
Secondary: Factuality Error Rate (%), Ingestion Fidelity (%)
11. Guardrails (Zero-Tolerance & Thresholds)
Zero-Tolerance: Unsupported claims, PII leaks
Thresholds: Ingestion Fidelity ≥ 98%
Latency +3s max; Cost +20% max
12. Experiment Design
Offline Paired Comparison (paired by doc & Q). Randomized order in judging.
13. Data / Sample
500 paired examples (min. 200 per slice). Stratified by doc type and length.
14. Decision Rule
Ship if Faithfulness lift ≥ +4 pts with 95% CI lower bound > 0 and no zero-tolerance failures.
15. Rollback Plan
Disable feature flag. Revert to baseline pipeline. Monitor for 24–48h.
16. Success Definition
Lift ≥ +4 pts in Faithfulness, no guardrail violations, and positive user feedback in canary.
Pre-registered by:
Asha Rao
Asha Rao, Lead Evaluator
May 7, 2025
Notes / Links / Attachments
Analysis plan (link) | Dataset card (link) | Judge prompt v1.2 (link) | Dashboard (link)

Learning Archive: Turning Experiments into Organizational Memory

A team that runs experiments without a Learning Archive will rediscover the same lessons every six months. The Archive is what compounds.

A Learning Archive entry should capture:

- What was tested?
- What was the hypothesis?
- What changed?
- What worked?
- What failed?
- Which slices benefited?
- Which guardrails were stressed?
- What should future teams avoid repeating?
- What new hypothesis emerged from the result?

Worked Archive Entry: E-2026-Q2-FIN-INGEST-01

A condensed example of the artifact filed after the experiment closes.

- **Tested:** Table-aware chunking + chart-structure extraction on scanned financial reports.
- **Hypothesis:** ≥4-pt Faithfulness lift and ≥5-pt Cross-Modal Consistency lift, no guardrail regression.
- **Result:** Faithfulness +5.2 pts (CI: +3.8 to +6.6). Cross-Modal Consistency +6.1 pts (CI: +4.5 to +7.7). Ingestion Fidelity regressed −1.9 pts on the dense-table slice due to caption-cell absorption.
- **Decision:** Investigate. Ramp blocked.
- **What worked:** The chart parser was the dominant contributor. Ablation showed the Cross-Modal Consistency lift came almost entirely from the parser, not the chunker.
- **What failed:** The table-aware chunker absorbed caption text into adjacent cells on documents with small typography. Affected approximately 12% of dense-table cases.
- **Slices benefited:** Chart-heavy reports (+8.4 pts CMC). Multilingual reports showed no measurable effect — flagged for follow-up.
- **Guardrails stressed:** Ingestion Fidelity on dense-table slice. Latency held (+180ms p95, inside budget).
- **Avoid repeating:** Do not couple two ingestion changes in a single experiment without explicit ablation. The regression was hidden by the headline lift for two days before slice analysis surfaced it.

- **New hypothesis emerging:** A caption-preservation post-processing step will eliminate the dense-table regression while retaining the Cross-Modal Consistency gain. Filed as E-2026-Q2-FIN-INGEST-02.

A failed experiment is not wasted if it enters the Learning Archive. It becomes part of the organization's map of what does not move quality.

The Archive connects naturally to EvalOps (it is part of the audit trail) and to the Risk-Aware Operating Loop (it feeds the "what happened" step with structured organizational memory rather than tribal recall).

Over time, the Archive becomes the most valuable evaluation artifact a team owns. Models change. Prompts change. Vendors change. The Archive — what we tried, what worked, what did not, and why — outlasts all of them.

How This Fits into the Risk-Aware Operating Loop

The Risk-Aware Operating Loop asks five questions in sequence:

1. What could go wrong?
2. What will we commit to?
3. What did we measure?
4. What will we do?
5. What happened?

Experimentation adds four questions that sit across the loop, not after it:

- What do we believe will improve the loop?
- How will we test that belief?
- What did we learn?
- What should change next?

Those four additions transform the loop. Without them, the loop is a governance system — auditable, defensible, slow. With them, the loop becomes a learning system — auditable, defensible, *and* compounding.

This is the strategic case for the experimentation layer. It is not a tooling upgrade. It is the difference between an organization that ships Multimodal AI products and an organization that learns to ship better Multimodal AI products faster than its competitors.

Closing: From Scorekeeping to Learning

Multimodal AI teams do not just need more evals. They need experimental discipline around the evals they already run.

A dashboard tells a team what moved. A statistical report tells them whether the movement is trustworthy. An experiment tells them why it moved, for whom it moved, what risks moved with it, and what to do next.

The disciplines compound.

Evaluation measures quality.

Statistics builds trust in measurement.

Experimentation creates learning.

EvalOps makes the learning repeatable.

Governance turns learning into decisions. Together they form the operating model of a Multimodal AI organization that improves on purpose rather than by accident.

Where to start

The minimum viable experimentation program needs three things, not thirty.

1. A CLAIM brief for the next release decision — even if it is half a page.
2. A pre-registered primary metric and at least one guardrail, written down *before* the experiment runs.
3. A Learning Archive entry filed within a week of the decision, including the experiments that did *not* work.

The rest — slice plans, judge calibration, Statistical Debt Registers, a deliberate experiment portfolio, the full Risk-Aware Operating Loop, agentic trajectory hooks — comes after the muscle is built. A team that does these three things consistently for a quarter will out-learn a team with a more sophisticated evaluation stack and no discipline.

If you are not sure where your team currently sits, use the Experimentation Maturity Model in §14 as a self-assessment. The most common stuck point is between Levels 2 and 3 — between writing down a hypothesis and writing down the slice plan that protects against the harm averages conceal. That single discipline is the one most worth building this quarter.

Hypothesis-driven Multimodal AI evaluation is the discipline of turning quality questions into testable beliefs, testable beliefs into calibrated evidence, and calibrated evidence into product learning.

The team that masters this discipline does not just ship better models. It builds the muscle to keep shipping better models, in domains and modalities its competitors cannot yet evaluate.

The future of Multimodal AI evaluation is not only better measurement. It is better learning.